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Roll No

CE-6001-CBGS

B.E. VI Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Design of Hydraulic Structures

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. What is meant by the elementary profile of a gravity dam; and how it is deduced? What should be the maximum depth of elementary profile of a dam if the safe limit of stress on the masonry should not exceed 1500 kN/m^2 ?
2. a) Differentiate between 'horizontal' and 'vertical' piping in earth dams. Suggest permanent measures to check vertical piping.
b) What are the causes of failure of earth dam? Explain.
3. Define the following terms and explain clearly;
a) Phreatic line in earth dams, and
b) Instability of a gravity dams.
4. a) Sketch an ogee spillway profile and mark it in the different zones.
b) Design a suitable section for the over flow section of a concrete gravity dam having the distance sloping at a slope of 0.7H; 1V. The design discharge for the spillway is 6000 cumecs. The height of the spillway above the river bed is 60m. The effective length of spillway may be taken as 50m.

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5. Write short notes on:
 - i) Volute siphon
 - ii) Side channel spillway or chute spillway or shaft spillway
 - iii) Spillway gate
 - iv) Energy dissipation below spillways
6. What are the principal components of a hydroelectric scheme?
Discuss utility of each component.
7.
 - a) What is surge tank and what are its types and uses?
 - b) What is draft tube and what are its uses?
8. Write short notes on any four of the following:
 - a) Tidal plants
 - b) Syphon aqueduct
 - c) Emergency spillway
 - d) Pore pressure
 - e) Galleries in gravity dam
